Internship Report: Encoding Spatial Information for Anomaly Detection Systems

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Abstract

Acoustic anomaly detection (ASD) is widely used in industrial monitoring to identify deviations from normal operating behavior based on sound. Although microphone arrays are increasingly employed in practice, most ASD research focuses on single-channel audio, leaving the role of spatial information underexplored. This internship investigates the contribution of spatial audio features to ASD and introduces a modular framework for systematic experimentation in an industrial setting.

A flexible ASD framework was implemented within Sorama's internal AI development platform, formalizing anomaly detection as the composition of feature extraction and anomaly scoring components. Using this framework, autoencoder-based ASD systems were evaluated on an in-house dataset recorded with a 64-channel microphone array. Spectral features based on mel spectrograms were compared with spatial features derived from generalized cross-correlation with phase transform (GCC-PHAT), as well as their combination.

Autoencoder-based systems consistently outperform Sorama's existing Isolation Forest baseline. Spectral features yield the strongest performance, while spatial features provide limited additional benefit under the evaluated conditions, highlighting challenges in exploiting spatial information for industrial ASD.

1 Introduction

1.1 Background and Motivation

Anomaly detection is a fundamental problem across a wide range of application domains, including industrial condition monitoring, medical diagnostics, and the development of robust machine learning systems capable of handling out-of-distribution inputs. In the acoustic domain, anomaly detection enables applications such as machine fault detection, machine condition monitoring, and health assessment, where deviations from normal sound patterns may indicate emerging failures or abnormal operating conditions.

Modern acoustic monitoring systems increasingly rely on microphone arrays rather than single microphones. By capturing sound at multiple spatially distributed positions, microphone arrays provide access to spatial information such as direction-of-arrival cues and spatial energy distributions. In principle, these spatial characteristics can complement conventional spectral and temporal representations of audio, potentially enabling more informative modeling of complex acoustic environments and improved detection of anomalous events.

1.2 Research Gap

Despite the widespread adoption of microphone arrays in practical acoustic monitoring systems, most existing research on acoustic anomaly detection focuses on single-channel audio representations. Significant progress has been made in learning effective spectral and temporal features from raw audio sig-

nals, yet comparatively little attention has been paid to the role of spatial information derived from multi-channel recordings. As a result, it remains unclear to what extent spatial audio representations can contribute to anomaly detection performance in realistic industrial settings. This gap motivates the present work, which investigates whether and how spatial cues obtained from microphone arrays can be incorporated into acoustic anomaly detection systems.

1.2.1 Industrial Context and Scope

This work was conducted as part of a five-month, part-time internship at Sorama (10 ECTS), a company specializing in acoustic monitoring solutions based on microphone array technology. Sorama employs an internal AI development platform, referred to as ADAM, which supports a range of machine learning models and provides preprocessing methods tailored to multi-channel acoustic data, including beamforming, acoustic heatmap computation, and sound source localization. Although there is an anomaly detection model implemented at Sorama already, there is currently no data-driven feature extraction model or a unified anomaly detection training and evaluation framework.

Given the limited duration and part-time nature of the internship, the scope of this work is primarily exploratory and architectural rather than exhaustive. The focus lies on enabling systematic experimentation with different anomaly detection strategies and input representations, rather than on optimizing a single model or conducting large-scale empirical studies.

1.3 Objectives and Contributions

The objectives of this internship are twofold. First, the work aims to investigate the effect of incorporating spatial information derived from microphone arrays into acoustic anomaly detection systems in an industrial context. Second, it seeks to design and implement a modular anomaly detection framework that facilitates systematic experimentation with different feature extraction methods and anomaly scoring approaches.

The main contributions of this work are as follows:

- the formalization of acoustic anomaly detection systems in a modular framework that explicitly separates feature extraction from anomaly scoring.
- the implementation of this framework within Sorama's internal AI development platform, ADAM.
- the development of a selection of autoencoder-based anomaly detection systems that serves as a reference implementation for evaluating different audio representations.

2 Background information

2.1 Data-Driven Anomalous Sound Detection

Acoustic Anomaly Detection (ASD) aims to identify sounds that deviate from a learned notion of normal operating behavior. A representative application is Machine Inspection (MI), in which machinery is continuously monitored and malfunctions are detected through anomalous acoustic patterns. Early detection of such anomalies is critical, as it can significantly reduce operational losses and enable timely maintenance. With the increasing interconnection of machinery in the context of Industry 4.0, the relevance and applicability of ASD have grown substantially.

Most ASD approaches focus on learning a representation of normal operating conditions rather than explicitly modeling anomalous behavior. This design choice is largely driven by practical constraints: anomalous data are typically scarce, heterogeneous, and difficult to characterize exhaustively, whereas normal operating data are more abundant and comparatively homogeneous. As a result, directly training models on anomalous samples is often infeasible in real-world industrial settings.

In practice, ASD systems can be broadly categorized into two main classes, as well as hybrids between them: generative models and discriminative models

[1]. Discriminative approaches are typically trained using synthetic anomalies or auxiliary classification tasks, while generative approaches aim to model the distribution of normal operating data directly.

From a broader perspective, both generative and discriminative approaches to acoustic anomaly detection can be understood as methods for learning feature representations that are well suited to the downstream binary classification task of distinguishing normal from anomalous sounds. While discriminative models are explicitly optimized to separate normal data from synthetic or auxiliary anomalies, and generative models implicitly define a notion of normality through density estimation or reconstruction, both paradigms ultimately rely on the quality of the learned feature space. Anomalies are detected not directly in the raw input domain, but as deviations in this learned representation.

2.2 Spatial Audio and Microphone Arrays

One key benefit of data captured by microphone arrays is the information implicit in the different times at which signals reach microphones in the array. These time differences of arrival (TDOA) arise from the spatial separation of microphones and the finite propagation speed of sound, and they encode information about the location and movement of sound sources within the monitored environment. In contrast to single-channel recordings, multi-channel audio therefore provides access not only to spectral and temporal characteristics of sound, but also to its spatial structure.

Spatial audio processing techniques aim to exploit this additional information to characterize the acoustic scene. Common approaches include beamforming, which enhances sound arriving from specific directions while suppressing others, and sound source localization methods that estimate the direction or position of dominant sound sources. Another widely used class of techniques derives spatial features directly from inter-channel relationships, for example through inter-channel phase or time-delay representations.

From a representation learning perspective, spatial features extracted from microphone arrays provide a complementary view of the acoustic signal compared to conventional single-channel spectral representations. While spectral features primarily describe the frequency content and temporal evolution of sound, spatial features encode how acoustic energy is distributed across space and how this distribution changes over time. In principle, such information may be particularly informative in scenarios where anomalous events are associated with changes in source position, radiation pattern, or spatial interference structure, even when spectral characteristics remain similar to those of normal operation.

3 System Design and Methodology

3.1 ASD System Formalism

To provide a structured framework for the development and evaluation of acoustic anomaly detection systems, we define a high-level abstraction of an ASD pipeline. Following previous work in high-level ASD evaluation tools [2], we formalize an ASD system as the composition of two mapping functions.

The first component is a *feature extractor* $f_\theta(x) : X \mapsto F$, which maps an input signal $x \in X$, representing raw audio, into a feature space F. The parameters θ denote the (possibly learned) parameters of the feature extraction process, which may range from handcrafted signal processing pipelines to deep neural network–based encoders.

The second component is an *anomaly scorer*, defined as $s(z) : F \mapsto [0, 1]$, which assigns a scalar score to a feature representation $z = f_\theta(x)$. This score quantifies the degree to which the input deviates from the learned notion of normality, with higher values indicating a higher likelihood of anomalous behavior.

In practice, the anomaly score is typically compared against a predefined or learned threshold $\tau \in [0, 1]$ to obtain a binary decision, i.e.,

$$\hat{y} = \begin{cases} \text{anomalous}, & s(z) > \tau, \\ \text{normal}, & \text{otherwise.} \end{cases}$$

The choice of threshold may depend on application-specific requirements, such as acceptable false positive rates or operational constraints, and is often determined using validation data.

This abstraction encompasses a wide range of ASD approaches. In generative models, such as autoencoders or density estimators, the anomaly scorer may be derived from reconstruction error or likelihood estimates. In discriminative approaches, the scoring function may correspond directly to a classifier output trained using auxiliary tasks or synthetic anomalies. Importantly, while these approaches differ in how the anomaly score is computed, both adhere to the same high-level structure of mapping raw audio to a feature space followed by anomaly scoring.

An advantage of this formalization is the explicit separation between feature extraction and anomaly scoring. This modular decomposition allows different feature representations and scoring mechanisms to be combined and evaluated independently, facilitating experimentation and fair comparison between ASD strategies.

3.2 Existing Anomaly Detection Approach at Sorama

Sorama's existing acoustic anomaly detection pipeline primarily relies on the Isolation Forest algorithm, a tree-based method that identifies anomalies by recursively partitioning the feature space using randomly constructed binary trees. Isolation Forests are attractive in industrial settings due to their computational efficiency, low memory footprint, and scalability to large datasets, and have demonstrated reasonable performance on Sorama's in-house data.

However, the effectiveness of Isolation Forests depends on a strong structural assumption: anomalous samples are expected to reside in low-density regions of the feature space and can therefore be isolated with relatively few splits. While this assumption often holds for tabular or carefully engineered features, it is poorly aligned with raw or minimally processed audio data. Acoustic signals are inherently high-dimensional and exhibit strong temporal and spectral structure, causing anomalies to manifest as complex pattern deviations rather than isolated points.

As a result, applying Isolation Forests to audio-based machine condition monitoring typically requires substantial hand-crafted feature engineering to produce representations compatible with the algorithm's assumptions. This process is application-specific and limits generality, motivating the exploration of representation learning–based approaches and a more modular anomaly detection framework.

3.3 ASD Framework Design

The main contribution of this internship is the development of a modular ASD framework that enables flexible, reproducible training and evaluation of anomaly detection pipelines within Sorama's AI development environment.

Sorama's existing workflow is centered around monolithic model classes instantiated through configuration files, with the `adam` command-line tool orchestrating preprocessing, training, and evaluation. While effective for end-to-end models, this setup is poorly suited to typical ASD pipelines, which consist of a feature extractor followed by an anomaly scorer and are rarely trained jointly. In practice, both components must be trained separately while sharing identical data splits, preprocessing steps, and intermediate representations. The proposed framework addresses this limitation by explicitly supporting composite pipelines and coordinating their sequential training and evaluation within a single execution.

Within the framework, users define a feature extractor and an anomaly scorer as independent components, each of which can be configured as trainable or frozen. Training proceeds sequentially: the feature extractor is trained first, if enabled, and subsequently used to generate feature representations for the training data. These representations are then used to train the anomaly scorer. The scorer may additionally consume raw inputs from the dataset, enabling hybrid approaches such as reconstruction-based methods that require access to both learned features and original signals. By managing both stages within a unified training loop, the framework ensures consistent data splits and eliminates the need for manual coordination across separate runs.

During validation and before evaluation, the framework optionally applies post-processing to the anomaly scores. Scores can be normalized using min-max scaling, after which a decision threshold is estimated using only normal validation data. Threshold estimation is performed by fitting a parametric distribution to the score distribution (currently a Gaussian or gamma distribution)—and defining anomalies as scores exceeding a specified percentile. This procedure aligns with standard ASD assumptions and avoids reliance on anomalous samples during calibration, although the framework can be easily adapted to facilitate this as well.

To support inference on variable-length audio inputs, the framework implements temporal windowing. Inputs longer than the temporal context supported by the model are segmented into fixed-length windows, which are processed independently by the feature extractor and anomaly scorer. The resulting window-level anomaly scores may be aggregated into a single clip-level score, such as by averaging. While this provides a robust global decision, the design also allows models to retain window-level scores for temporally localized anomaly detection.

For deployment, the trained feature extractor, anomaly scorer, and associated preprocessing steps (including windowing and score aggregation) are composed into a single saveable and loadable TensorFlow Keras model, enabling streamlined inference through a unified model interface rather than requiring separate configuration of individual pipeline components.

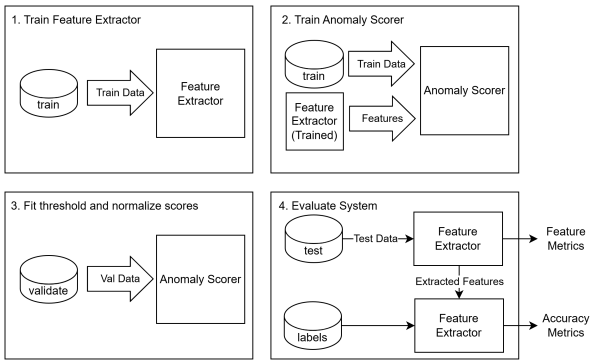

Figure 1: Overview of the proposed training and evaluation architecture for the anomaly detection framework. The pipeline consists of four stages. (1) A feature extractor is trained using normal training data. (2) The trained feature extractor is used to generate feature representations, which are then employed to train an anomaly scorer on the training set. (3) Validation data are used to fit score normalization parameters and determine the anomaly detection threshold. (4) During evaluation, test data are passed through the trained feature extractor and anomaly scorer to produce anomaly scores, which are compared against ground-truth labels to assess system performance.

4 Experimental Setup

To evaluate the proposed modular ASD framework and investigate the effect of different feature representations, a set of controlled experiments was conducted on an in-house industrial anomaly detection dataset. Autoencoder-based ASD systems were trained using two distinct feature representations: mel spectrograms and spatial features derived from generalized cross-correlation with phase transform (GCC-PHAT). In addition, a combined representation using both feature types was evaluated.

The resulting systems were compared using multiple anomaly scoring methods, and a baseline system based on Sorama's existing Isolation Forest approach with hand-engineered wavelet features was included for reference.

4.1 Cargill Dataset and Evaluation Metrics

The experiments were performed using an in-house dataset collected at Cargill facilities in collaboration with Boston Dynamics. Acoustic measurements were acquired using Sorama's CAMiV64 microphone array device [3] during multiple site visits. Recordings were obtained at varying distances and viewing angles relative to the monitored asset, while ensuring that the asset remained centrally located with respect to the array. The dataset contains a limited set of industrial assets, including motors, bearings, and fans, which were labeled by on-site engineers based on maintenance assessments.

Anomaly detection performance was evaluated using the area under the receiver operating characteristic curve (ROC–AUC). This metric is widely adopted in the acoustic anomaly detection literature and has been used previously within Sorama to assess ASD system performance. ROC–AUC provides a threshold-independent measure of detection capability and reflects the combined effect of feature extraction and anomaly scoring.

In addition to quantitative evaluation, the structure of the learned feature representations was examined qualitatively using dimensionality reduction techniques. Principal component analysis (PCA), t-distributed stochastic neighbor embedding (t-SNE), and uniform manifold approximation and projection (UMAP) were used to visualize embeddings and assess clustering behavior and separability between normal and anomalous samples.

4.2 Autoencoder-Based Anomaly Detection

Autoencoders constitute one of the most commonly used model classes for anomaly detection. They are self-supervised models trained to reconstruct their input after passing it through a bottleneck layer, thereby learning a compact latent representation of the data. An autoencoder consists of an encoder network, which maps the input to a lower-dimensional latent space, and a decoder network, which reconstructs the input from this representation by minimizing reconstruction error.

In the present work, a fully connected autoencoder architecture was employed. The input dimensionality is feature-dependent and determined by the chosen feature representation. The encoder consists of a sequence of symmetric hidden layers with sizes 128, 128, and 64, followed by a latent bottleneck layer of dimension 32. The decoder mirrors this structure in reverse, reconstructing the input from the latent representation. This architecture was selected as a lightweight yet expressive baseline that is compatible with both spectral and spatial feature representations.

In acoustic anomaly detection, a common anomaly scoring strategy is based on reconstruction error, under the assumption that an autoencoder trained exclusively on normal operating data will accurately reconstruct such inputs while producing larger errors for anomalous samples. In addition to this decoder-based scoring approach, multiple alternative anomaly scorers were implemented to evaluate the quality of the learned latent representations and to demonstrate the modularity of the proposed framework.

Specifically, the following anomaly scoring methods were applied to the autoencoder latent features:

- **k-Nearest Neighbours (kNN)** with $k = 20$, using the average distance to the k nearest neighbors as the anomaly score;
- **Local Outlier Factor (LOF)** with $k = 20$, capturing local density deviations in the latent space;
- **Mahalanobis distance**, modeling the latent feature distribution under a multivariate Gaussian assumption;
- **Decoder reconstruction error**, computed as the normalized discrepancy between the input and its reconstruction.

Relating this setup to the ASD system formalism introduced in Section 3.1, the encoder implements the feature extraction function f_θ, while each anomaly scorer defines a distinct instantiation of the scoring function $s(\cdot)$. By decoupling representation learning from anomaly scoring, the framework enables systematic evaluation of different scoring strategies operating on identical learned features.

4.3 Spectral and Spatial Feature Representations

All models were trained on the Cargill Raw dataset using feature representations derived from the original multi-channel acoustic recordings. Two distinct input feature types were considered: spectral features based on mel spectrograms and spatial features based on GCC-PHAT.

For the spectral representation, mel spectrograms were computed after applying beamforming in the forward (straight) direction to the raw microphone array data, yielding a single-channel audio signal. The beamformed signal was segmented into non-overlapping frames, and a short-time Fourier transform (STFT) was applied using an FFT size of 4096. No overlap between consecutive frames was used. The resulting power spectra were mapped onto the Mel scale to obtain mel spectrogram features. To incorporate limited temporal context, five consecutive mel spectrogram frames were concatenated along the feature dimension and used as a single input sample to the model.

Spatial features were computed directly from the raw multi-channel recordings using GCC-PHAT. This representation captures inter-channel time-delay information and scales with the number of microphones in the array. As the number of microphone pairs grows quadratically with the array size, direct computation of all pairwise cross-correlations would result in prohibitively high-dimensional feature vectors. To mitigate this, dimensionality reduction was applied by restricting the computation of cross-correlations to microphone pairs within predefined k-nearest-neighbor (kNN) neighborhoods on the array geometry. For each selected microphone pair, GCC-PHAT was computed using five discrete time lags, resulting in a compact spatial feature representation.

Both feature representations were used as inputs to the autoencoder architectures described in the previous subsection. The dimensionality of the resulting input vectors therefore differed between spectral and spatial features, but all representations were processed using the same downstream framework and anomaly scoring methods to enable fair comparison.

5 Results

This section presents the experimental results obtained using the proposed modular ASD framework on the Cargill dataset. Performance is reported for autoencoder-based systems trained on different feature representations and evaluated using multiple anomaly scoring methods, as described in the previous section. In addition, results are compared against Sorama's existing Isolation Forest–based baseline.

Table 5 summarizes the ROC–AUC scores achieved by the evaluated anomaly scoring methods when applied to latent representations learned from mel spectrogram features, GCC-PHAT features, and their combination. For autoencoder-based systems, results are reported for k-nearest neighbors (kNN), local outlier factor (LOF), Mahalanobis distance, and decoder reconstruction error scoring. All scores are computed at the clip level following the windowing and aggregation procedure described in Section 4.

Across feature representations, decoder-based reconstruction error scoring consistently achieves the highest ROC–AUC values among the evaluated anomaly scoring methods. For mel spectrogram features, the autoencoder with decoder-based scoring attains the strongest overall performance, while alternative scorers operating in latent space exhibit lower but comparable performance. When trained on GCC-PHAT features, autoencoder-based systems achieve competitive reconstruction-based performance, although distance- and density-based scoring methods exhibit reduced effectiveness.

The combined feature representation, obtained by concatenating latent features learned from mel spectrogram and GCC-PHAT inputs, does not yield a consistent improvement in ROC–AUC across the

evaluated scoring methods. In this configuration, performance remains comparable to or below that of systems trained on mel spectrogram features alone.

5.1 Comparison with Isolation Forest Baseline

To assess the practical impact of the proposed framework, the performance of the autoencoder-based ASD systems was compared against Sorama's existing Isolation Forest–based anomaly detection approach using hand-engineered wavelet features. The Isolation Forest baseline achieves a ROC–AUC of 0.735 on the Cargill dataset.

All evaluated autoencoder-based systems outperform the Isolation Forest baseline when using mel spectrogram features and decoder reconstruction error scoring. In particular, the autoencoder trained on mel spectrogram inputs achieves a substantially higher ROC–AUC, indicating improved discrimination between normal and anomalous operating conditions. Several latent-space scoring methods, including k-nearest neighbors and Mahalanobis distance, also exceed the baseline performance, although to a lesser extent.

When trained on GCC-PHAT features, autoencoder-based systems continue to outperform the Isolation Forest baseline under reconstruction-based scoring, while alternative scoring methods exhibit more variable performance. Overall, these results demonstrate that the proposed framework enables anomaly detection systems that achieve higher performance than Sorama's current production-oriented baseline on the evaluated dataset.

6 Discussion

The results presented in the previous section demonstrate that the proposed modular ASD framework is fully functional and suitable for systematic experimentation with different feature representations and anomaly scoring strategies. The framework was successfully implemented, tested, and applied to a real-world industrial dataset, serving as a proof of concept for both research-oriented evaluation and deployment-oriented development within Sorama's AI platform.

6.1 Autoencoder-Based Models versus Isolation Forest

Across the evaluated configurations, autoencoder-based anomaly detection systems consistently outperform Sorama's existing Isolation Forest baseline. This performance improvement is observed for both spectral and spatial input representations, indicating that representation learning–based approaches provide a more effective modeling of normal operating behavior than hand-engineered features combined with density-based isolation methods. These findings support the design choice to shift emphasis from manual feature engineering toward learned representations within the ASD pipeline.

In particular, autoencoders trained on mel spectrogram inputs achieve strong performance, highlighting the effectiveness of learned spectral representations for acoustic anomaly detection in the evaluated industrial setting. Furthermore, anomaly scoring methods operating on the learned latent space, such as k-nearest neighbors and Mahalanobis distance, are expected to exhibit more competitive performance relative to decoder reconstruction error. This suggests that lightweight, post-hoc scoring mechanisms may be sufficient when combined with expressive learned features, offering a favorable trade-off between detection performance and computational complexity.

6.2 Observations on GCC-PHAT-Based Features

One notable and initially surprising result is the relatively strong anomaly detection performance obtained when using decoder reconstruction error on autoencoders trained with GCC-PHAT features. Given that GCC-PHAT is intended to encode spatial information while suppressing spectral magnitude effects, such performance would not be expected to match that of mel spectrogram–based representations under typical assumptions.

Further analysis and discussions with engineers at Sorama suggest that this behavior may be influenced

Feature set	kNN	LOF	Mahalanobis	Decoder
Mel spectrogram autoencoder	0.615	0.731	0.721	0.826
GCC-PHAT autoencoder	0.593	0.610	0.523	0.821
Mel AE + GCC-PHAT AE	0.542	0.551	0.538	-

Table 1: ROC–AUC scores for different anomaly scoring methods across feature representations. Performance is reported for k-nearest neighbors (kNN), local outlier factor (LOF), Mahalanobis distance, and decoder reconstruction error, evaluated on autoencoder-based mel spectrogram features, GCC-PHAT features, and their combination.

by limitations in the dataset and feature construction. Specifically, restricting GCC-PHAT computation to small microphone neighborhoods emphasizes narrow-band correlations while underrepresenting wide-band spatial relationships. As a result, the resulting representation may exhibit implicit sensitivity to changes in specific frequency bands, despite GCC-PHAT being designed as a primarily spatial feature. Similar narrow-band filtering experiments conducted previously on the Cargill dataset have yielded reasonable anomaly detection performance, supporting this interpretation.

These observations indicate that the observed performance of GCC-PHAT–based systems may not be attributable solely to spatial information, but rather to unintended spectral sensitivity introduced by the dimensionality reduction strategy. This highlights the importance of carefully designing spatial feature extraction and compression techniques to avoid bias toward specific frequency components when the goal is to isolate spatial cues.

7 Implications and Future Directions

The limited benefit observed from combining mel spectrogram and GCC-PHAT features suggests that joint spectral–spatial feature spaces in the current setup do not yet yield strongly complementary representations. This may be attributed to several factors, including limited spatial variability in the dataset, overly aggressive dimensionality reduction applied to the microphone array data, or architectural constraints in how spectral and spatial features are fused. Future work should therefore focus on developing dimensionality reduction methods for high-channel-count microphone arrays that preserve broadband spatial characteristics without introducing unintended spectral bias. Potential directions include selecting a subset of the 64 microphones that has a specified geometry or learning other geometry-aware spatial features. Exploring alternative spatial–spectral feature representations, such as SALSA or related methods, may also provide a more balanced integration of spectral and spatial information within a unified representation. In addition to feature-level fusion, end-to-end learning approaches that operate directly on fused multi-channel representations present a promising direction. Rather than relying on hand-crafted spatial features, autoencoder-based models could be extended to learn joint spectral–spatial embeddings directly from raw or minimally processed array data, provided that sufficient training data and appropriate regularization strategies are available. At the task level, the formulation of anomaly detection as a purely clip-level classification problem may inherently limit the utility of spatial information. Future research could therefore explore spatially informed objectives, such as sound event localization and detection (SELD), which explicitly incorporate source location information and may encourage the learning of more meaningful spatial representations. Related approaches based on spatial discretization via beamforming are also of interest. Jowever, initial experiments in this direction indicated elevated anomaly scores in spatial regions with low acoustic energy, suggesting that such meth-

ods require improved training coverage and more robust handling of out-of-distribution signals when there is no asset being beamformed towards. Overall, while the present results indicate that spectral features dominate anomaly detection performance under the evaluated conditions, they do not preclude the usefulness of spatial information in more diverse or spatially dynamic settings. The modular framework developed in this work provides a flexible foundation for systematically exploring these directions and supports future extensions toward more spatially aware industrial ASD systems.

8 Conclusion

This internship investigated the role of spatial audio information in acoustic anomaly detection and resulted in the design and implementation of a modular ASD framework within Sorama's internal AI development platform. The framework formalizes anomaly detection as the composition of feature extraction and anomaly scoring components, enabling flexible, reproducible experimentation across different representations and detection strategies while supporting streamlined deployment.

Using this framework, autoencoder-based anomaly detection systems were prototyped and evaluated on an industrial dataset. These systems consistently outperformed Sorama's existing Isolation Forest–based approach, demonstrating the practical value of representation learning for acoustic anomaly detection in real-world settings. Learned mel spectrogram representations proved particularly effective, while lightweight post-hoc anomaly scorers operating on latent features achieved competitive performance.

Although the incorporation of spatial features did not yield substantial performance gains under the evaluated conditions, the results provide important insight into the limitations imposed by dataset characteristics and feature construction choices. Rather than indicating a lack of potential for spatial audio, these findings emphasize the need for more suitable representations, tasks, and data to fully exploit spatial information.

Overall, the work delivers both a deployable anomaly detection framework and a structured experimental foundation for continued research. It provides Sorama with a practical tool for developing and evaluating future ASD systems and establishes a basis for deeper investigation into spatially informed acoustic anomaly detection.

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